

NHL — T-Cell Underground (subway system)



1. T tunnel header

WHAT YOU SEE

Big 'T' at tunnel mouth

WHAT IT ENCODES

Mature (peripheral) T-cell and NK-cell lymphomas are a minority of NHL — only ~10-15% in Western countries (higher in Asia where HTLV-1 and EBV-driven types cluster). As a group they carry a WORSE prognosis than the common aggressive B-cell lymphomas because they respond poorly to CHOP-type chemo and lack a rituximab equivalent (no CD20). The big exceptions to the 'T-cell = bad' rule are ALK-POSITIVE anaplastic large cell lymphoma and most cutaneous mycosis fungoides, which are indolent/curable. Mnemonic for the subway stops: 'Please Always Avoid Awful Nasty MF' = PTCL, ALCL, AITL, ATLL, NK/T-nasal, MF/Sézary.

2. Station 1: PTCL-NOS

WHAT YOU SEE

'STATION 1 PTCL-NOS'

WHAT IT ENCODES

Peripheral T-cell lymphoma, not otherwise specified (PTCL-NOS) is the single most common mature T-cell lymphoma (~25-30% of PTCL) and is a diagnosis of exclusion once specific entities are ruled out. It is aggressive, presents in older adults with widespread nodal disease, B symptoms, eosinophilia and pruritus, and frequently shows an aberrant T-cell immunophenotype with loss of a pan-T marker (CD5 or CD7). First-line is CHOP/CHOEP, but 5-year overall survival is only ~30%, and consolidation with autologous stem-cell transplant in first remission is often pursued.

BOARD TRAP

WRONG to reach for rituximab-CHOP (R-CHOP) when you see PTCL — these are T-cell tumors that are CD20-NEGATIVE, so rituximab does nothing. The discriminator: confirm a T-cell phenotype (CD2/CD3/CD5/CD7, often with aberrant antigen loss) before choosing therapy; the 'R' is only added for B-cell (CD20+) lymphomas.

3. PTCL 30% poor

WHAT YOU SEE

'POOR' sign with '30%'

WHAT IT ENCODES

The '30%' nails the dismal long-term outlook: 5-year overall survival for most peripheral T-cell lymphomas is roughly 25-35%, markedly inferior to DLBCL (~60-70% cured with R-CHOP). Drivers of the poor outcome are chemoresistance, late presentation, and high International Prognostic Index scores. This is why frontline CHOEP (CHOP + etoposide) and autologous transplant consolidation are favored in fit patients, in contrast to aggressive B-cell disease where R-CHOP alone often suffices.

BOARD TRAP

WRONG to assume a young, fit T-cell-lymphoma patient should be 'watched' or treated like an indolent lymphoma. The discriminator: PTCL-NOS, AITL, and ALK-negative ALCL are aggressive — delay costs cures; they need prompt anthracycline-based therapy ± transplant, unlike the indolent cutaneous T-cell lymphomas (mycosis fungoides).

4. Station 2: ALCL ALK+

WHAT YOU SEE

'STATION 2 ALCL' green ALK+

WHAT IT ENCODES

Anaplastic large cell lymphoma is a CD30-POSITIVE (uniformly, strong) T-cell lymphoma defined by large 'hallmark cells' with horseshoe/kidney-shaped nuclei. The ALK-POSITIVE subtype carries the t(2;5)(p23;q35) NPM1-ALK fusion, occurs in children and young adults, and has an excellent prognosis (5-year survival ~70-80%). Frontline therapy is now brentuximab vedotin + CHP (cyclophosphamide-doxorubicin-prednisone, replacing vincristine), which beat CHOP in the ECHELON-2 trial for CD30+ PTCL.

BOARD TRAP

WRONG to lump ALK+ ALCL with the other 'bad' T-cell lymphomas and give an aggressive transplant-in-first-remission approach. The discriminator: ALK POSITIVITY (and young age) flags a FAVORABLE tumor curable with chemo alone — reserve transplant for relapse; it is the ALK-NEGATIVE and older patients who do poorly.

5. ALK- ALCL 40-50%

WHAT YOU SEE

Red 'ALK-' with '40-50%'

WHAT IT ENCODES

ALK-NEGATIVE ALCL occurs in older adults, still CD30+, but lacks the ALK fusion and has an intermediate prognosis (~40-50% 5-year survival) — worse than ALK+ but better than PTCL-NOS; DUSP22-rearranged ALK-neg cases behave more favorably while TP63-rearranged ones do worse. A separate, board-favorite entity is BREAST-IMPLANT-ASSOCIATED ALCL: arises in the periprosthetic fluid/capsule around TEXTURED implants, usually presents as a late seroma, is ALK-negative/CD30+, and is often cured by implant removal + capsulectomy alone.

BOARD TRAP

WRONG to manage a periprosthetic seroma around a breast implant as benign or to start systemic chemo immediately. The discriminator: send the seroma fluid for cytology/CD30 — breast-implant-associated ALCL is usually localized and cured surgically (en-bloc capsulectomy), so reflexive chemotherapy overtreats early-stage disease.

6. CD30 / brentuximab link

WHAT YOU SEE

Green ALK bolt figure (CD30+ ALCL)

WHAT IT ENCODES

CD30 (Ki-1) is uniformly and strongly expressed in ALCL and is the therapeutic target of brentuximab vedotin, an antibody-drug conjugate that links an anti-CD30 antibody to the antimitotic toxin MMAE (monomethyl auristatin E). In the ECHELON-2 trial, brentuximab + CHP became frontline for CD30+ PTCL; the same drug treats relapsed Hodgkin lymphoma (also CD30+). Its signature toxicity is dose-dependent PERIPHERAL NEUROPATHY from the MMAE payload.

BOARD TRAP

WRONG to attribute brentuximab's neuropathy to the antibody or to keep dosing through worsening numbness. The discriminator: the MMAE microtubule-poison payload (like vincristine) causes the sensory neuropathy — that's also why vincristine is dropped from CHP when brentuximab is added, to avoid stacking neurotoxicity.

7. Station 3: AITL

WHAT YOU SEE

'STATION 3 AITL — FOLLICULAR HELPER'

WHAT IT ENCODES

Angioimmunoblastic T-cell lymphoma arises from T-FOLLICULAR-HELPER (Tfh) cells, marked by CD4, CD10, PD-1, CXCL13, BCL6 and ICOS, with recurrent IDH2, TET2, DNMT3A and RHOA mutations. Histology shows a polymorphous infiltrate with prominent arborizing high endothelial venules and expanded follicular dendritic cell networks, and EBV-positive bystander B-cells that can transform into a secondary DLBCL. It is aggressive (5-year survival ~30%) and notorious for autoimmune/inflammatory features that mimic infection or a connective-tissue disease.

BOARD TRAP

WRONG to misdiagnose an older patient with fevers, rash, generalized lymphadenopathy and a positive Coombs as a drug reaction, viral syndrome, or autoimmune disease. The discriminator: AITL produces these immune phenomena (polyclonal hypergammaglobulinemia, autoimmune hemolytic anemia) — biopsy a node showing Tfh markers (CD10/PD-1/CXCL13) and arborizing venules instead of treating it as benign.

8. AITL autoimmune antibody

WHAT YOU SEE

'autoimmune antibody' label

WHAT IT ENCODES

The immune dysregulation of AITL is a classic exam hook: polyclonal HYPERGAMMAGLOBULINEMIA, COOMBS-positive autoimmune hemolytic anemia, cold agglutinins, circulating immune complexes, rash/pruritus, polyarthritis, and generalized lymphadenopathy with hepatosplenomegaly. Labs often show elevated LDH, eosinophilia, and a positive rheumatoid factor or ANA, mimicking lupus or a viral illness. These features stem from the malignant Tfh cells driving an exuberant B-cell/plasma-cell response (hence the EBV+ B-cell expansion).

BOARD TRAP

WRONG to interpret the polyclonal hypergammaglobulinemia as a plasma-cell disorder (myeloma) or a benign reactive process. The discriminator: AITL gives a POLYCLONAL gammopathy (broad-based on SPEP) with systemic autoimmune features, not the MONOCLONAL spike of myeloma — and the underlying disease is a T-cell lymphoma diagnosed on lymph-node biopsy.

9. Station 4: ATLL

WHAT YOU SEE

'STATION 4 ATLL' acute/lymphomatous/chronic/smoldering

WHAT IT ENCODES

Adult T-cell leukemia/lymphoma is caused by HTLV-1 (human T-lymphotropic virus 1), a retrovirus transmitted via breastfeeding, sex, and blood/IV drug use, endemic to southwestern JAPAN, the CARIBBEAN, and parts of West Africa. The tumor cells are CD4+/CD25+ mature T-cells with a characteristic lobulated 'FLOWER CELL' nucleus on the smear. It has four clinical variants — ACUTE (most common, leukemic, very aggressive), LYMPHOMATOUS, CHRONIC, and SMOLDERING — with the acute/lymphomatous forms having median survival under a year.

BOARD TRAP

WRONG to confuse HTLV-1 with HIV or to assume it causes AIDS-type opportunistic-infection lymphomas. The discriminator: HTLV-1 is a transforming RETROVIRUS that drives a specific T-cell malignancy (ATLL) plus HTLV-1-associated myelopathy/tropical spastic paraparesis — it is NOT HTLV-3/HIV; EBV/HHV-8, not HTLV-1, drive the AIDS-associated lymphomas.

10. ATLL hypercalcemia

WHAT YOU SEE

Skull + Japan map at ATLL station

WHAT IT ENCODES

ATLL is the classic lymphoma cause of HYPERCALCEMIA, present in up to ~70% of acute cases, driven by tumor secretion of PTHrP (parathyroid-hormone-related peptide) and osteoclast-activating cytokines (RANKL, IL-1), producing LYTIC bone lesions. Patients are also profoundly immunosuppressed and prone to Strongyloides hyperinfection and PCP. Acute treatment of the hypercalcemia is IV fluids + bisphosphonate (zoledronic acid) ± calcitonin; the lymphoma itself is treated with antiviral therapy (zidovudine + interferon-α) for leukemic forms or intensive chemo.

BOARD TRAP

WRONG to attribute the hypercalcemia in ATLL to bone metastases or to check a PTH level and stop there. The discriminator: the mechanism is PTHrP-mediated (and osteoclast-activating cytokines), so PTH is SUPPRESSED while PTHrP is high — the board pattern is 'HTLV-1 + hypercalcemia + lytic lesions + flower cells = ATLL,' not metastatic bone disease.

11. Station 5: NK/T nasal

WHAT YOU SEE

'STATION 5 NK/T NASAL — SMILE'

WHAT IT ENCODES

Extranodal NK/T-cell lymphoma, nasal type is an EBV-DRIVEN, angioinvasive/angiodestructive lymphoma that classically produces a MIDLINE destructive lesion of the nose/palate/sinuses ('lethal midline granuloma'). Tumor cells are CD56+, cytoplasmic CD3ε+, and EBER-positive (EBV in situ). It is more common in Asia and Latin America. Because the cells express P-glycoprotein and resist anthracyclines, treatment uses non-anthracycline, asparaginase-containing regimens such as SMILE (steroid, methotrexate, ifosfamide, L-asparaginase, etoposide) with radiotherapy for localized disease.

BOARD TRAP

WRONG to work up a destructive midline nasal lesion solely as GPA (Wegener granulomatosis) or invasive fungal infection and miss the lymphoma. The discriminator: GPA is c-ANCA/PR3-positive with granulomatous vasculitis, whereas NK/T-nasal lymphoma is EBER-positive, CD56+, angiodestructive on biopsy — always send the nasal biopsy for EBER and CD56 before settling on a vasculitis diagnosis.

12. NK/T asparaginase

WHAT YOU SEE

'L-ASP' bottle at NK/T station

WHAT IT ENCODES

NK/T nasal lymphoma is uniquely ASPARAGINASE-SENSITIVE because the tumor cells cannot synthesize asparagine; L-asparaginase depletes circulating asparagine and starves them. Anthracycline (CHOP) regimens fail because the cells express multidrug-resistance P-glycoprotein, so asparaginase-based protocols (SMILE, AspaMetDex) are standard. Watch for asparaginase toxicities: pancreatitis, hyperglycemia, hepatotoxicity, hypofibrinogenemia/thrombosis, and hypersensitivity.

BOARD TRAP

WRONG to give standard CHOP first-line for NK/T nasal lymphoma. The discriminator: P-glycoprotein/multidrug resistance makes these tumors anthracycline-resistant — the cure comes from L-asparaginase-based regimens (SMILE) plus radiotherapy, which is also why asparaginase is a board buzzword tied to both NK/T lymphoma and ALL.

13. Station 6: MF / Sézary

WHAT YOU SEE

'STATION 6 MF/SEZARY'

WHAT IT ENCODES

Mycosis fungoides is the most common cutaneous T-cell lymphoma — an INDOLENT clonal proliferation of skin-homing CD4+ T-cells (CD3+/CD4+ with loss of CD7) that progresses slowly through PATCH → PLAQUE → TUMOR stages over years. Histology shows epidermotropic atypical lymphocytes forming PAUTRIER MICROABSCESSES (intraepidermal clusters) with cerebriform (convoluted) nuclei. Early skin-limited disease is treated topically (corticosteroids, nitrogen mustard, phototherapy/PUVA, local radiation); advanced disease uses bexarotene, interferon, romidepsin, or brentuximab.

BOARD TRAP

WRONG to dismiss chronic, treatment-refractory 'eczema' or 'psoriasis' that won't clear and to never biopsy it. The discriminator: persistent, atypical, biopsy-resistant skin patches in an adult with PAUTRIER MICROABSCESSES and a clonal CD4+/CD7-negative T-cell population is mycosis fungoides, not a benign dermatitis — repeat/deeper skin biopsies clinch it.

14. Sézary triad

WHAT YOU SEE

Red 'SEZARY' figure, erythroderma

WHAT IT ENCODES

Sézary syndrome is the leukemic/erythrodermic variant of cutaneous T-cell lymphoma, defined by the triad of (1) diffuse ERYTHRODERMA (often >80% body surface, with pruritus, alopecia, and 'leonine facies'), (2) generalized LYMPHADENOPATHY, and (3) circulating SÉZARY CELLS — clonal CD4+ T-cells with CEREBRIFORM nuclei (>1000/μL or elevated CD4:CD8 ratio >10). It is aggressive compared with patch-stage mycosis fungoides; treatment includes extracorporeal photopheresis, bexarotene, interferon, mogamulizumab (anti-CCR4), and romidepsin.

BOARD TRAP

WRONG to attribute new-onset erythroderma in an older adult purely to a drug reaction, psoriasis, or eczema. The discriminator: erythroderma + lymphadenopathy + circulating cerebriform Sézary cells (high CD4:CD8, CD7 loss) = Sézary syndrome — check a peripheral-blood flow cytometry and Sézary cell count rather than treating it as benign erythroderma.

15. EATL

WHAT YOU SEE

'EATL' tag

WHAT IT ENCODES

Enteropathy-associated T-cell lymphoma is an aggressive intestinal T-cell lymphoma arising as a complication of long-standing, refractory CELIAC disease (it is HLA-DQ2/DQ8 associated). It typically affects the proximal small bowel (jejunum) of middle-aged adults and presents with abdominal pain, weight loss, and sometimes bowel PERFORATION or obstruction. Outcomes are poor; treatment is surgery for complications plus chemotherapy (often CHOP-based ± transplant), but a strict gluten-free diet is protective against developing it.

BOARD TRAP

WRONG to attribute new abdominal pain, weight loss, or perforation in a known celiac patient to a celiac 'flare' or dietary noncompliance alone. The discriminator: refractory celiac disease with systemic symptoms, fevers, or an acute abdomen should raise suspicion for EATL — image the small bowel and biopsy, because untreated lymphoma masquerading as 'refractory sprue' is rapidly fatal.